

# Computational Physics 2 Classes No.9 - Report

## Advection-diffusion equation (FEM version)

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## 1 Introduction

The aim of this project is to solve the two-dimensional advection–diffusion equation using the finite element method (FEM) and periodic boundary conditions. The equation describes the transport of a scalar quantity under the combined action of advection, caused by a constant velocity field, and diffusion, characterized by a constant diffusion coefficient.

The computational domain is a square region of side length  $L = 5$ , and the time evolution is calculated using the Crank–Nicolson method. The initial condition is a Gaussian packet centered in the middle of the domain. Due to the periodic boundary conditions, any quantity leaving one side of the computational box re-enters through the opposite side.

Three physical situations are investigated. First, pure advection is considered in order to examine the transport of the packet and the preservation of its shape. Second, pure diffusion is analyzed by studying the evolution of the minimum and maximum values of the solution. Finally, the combined advection–diffusion process is investigated to determine whether the packet center moves with the velocity imposed by the velocity field.

## 2 Results

### 2.1 Pure advection

The first simulation was performed for pure advection with parameters  $D = 0$  and  $v_x = v_y = 1$ . Due to the periodic boundary conditions, the packet is transported across the computational domain and re-enters through the opposite boundaries after crossing the edges of the box.

Figure 1 shows the trajectory of the packet center. The center moves along the diagonal direction, which is consistent with the velocity field  $(1, 1)$ . Since the domain size is  $L = 5$ , the packet returns to its initial position after one period  $T = 5$  and reaches the same position again at  $t = 10$ , corresponding to the second return required in the assignment.

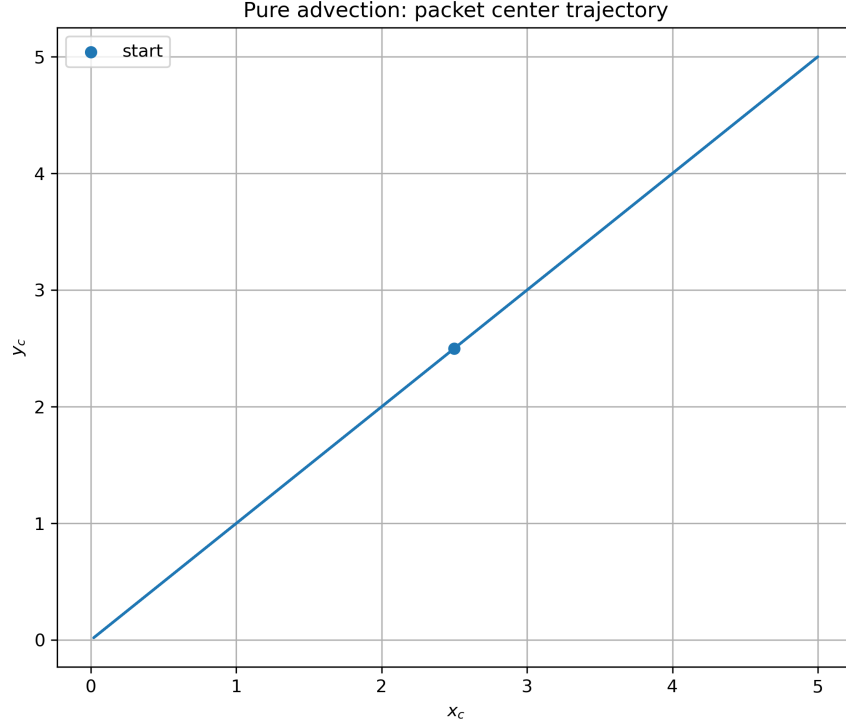


Figure 1: Trajectory of the packet center for pure advection.

To examine whether the packet shape remains unchanged during transport, the relative shape error was calculated with respect to the exact translated initial profile. The result is shown in Figure 2. The error increases slowly with time and remains below approximately 3% even after the second return of the packet. Therefore, the packet shape is nearly preserved and only a small numerical diffusion is observed.

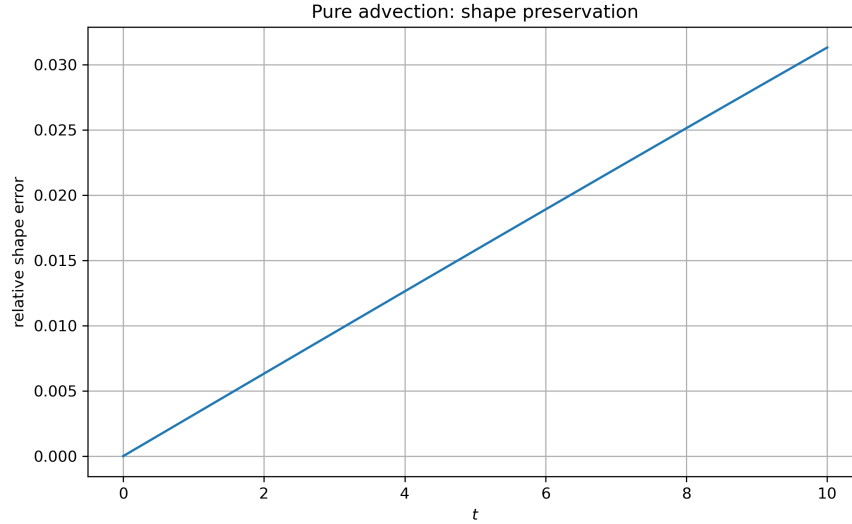


Figure 2: Relative shape error for pure advection.

## 2.2 Pure diffusion

The second simulation considered pure diffusion with parameters  $D = 0.1$  and  $v_x = v_y = 0$ . In this case the initial Gaussian packet spreads over the computational domain due to the diffusive process.

Figure 3 presents the minimum and maximum values of the solution as functions of time. The maximum value decreases monotonically from its initial value, while the minimum value increases from zero. This behavior is expected for diffusion, which tends to smooth spatial gradients and drive the system toward a uniform distribution.

At long times the difference between the minimum and maximum values becomes significantly smaller, indicating that the concentration profile approaches a nearly constant state throughout the domain.

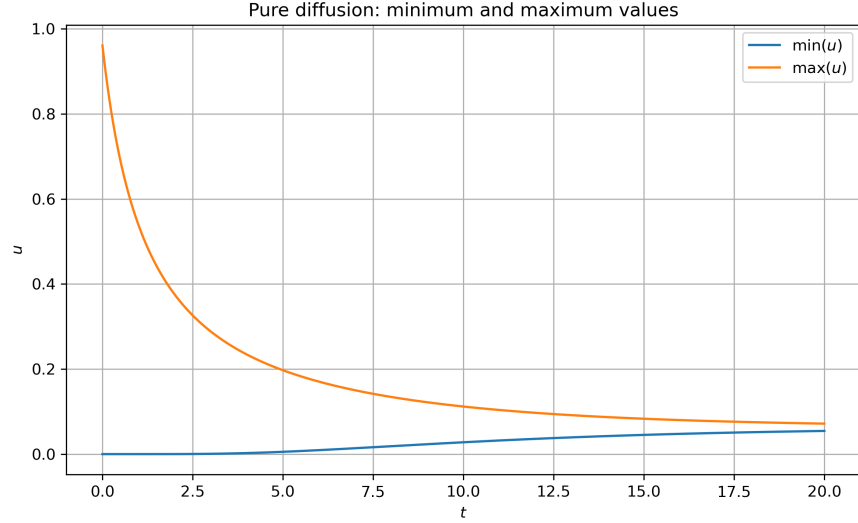


Figure 3: Minimum and maximum values of the solution for pure diffusion.

### 2.3 Advection–diffusion

The final simulation was performed for combined advection and diffusion with parameters  $D = 0.1$  and  $v_x = v_y = 1$ . In this case the packet is transported by the velocity field while simultaneously spreading due to diffusion.

Figure 4 shows the trajectory of the packet center. The center moves along the diagonal direction of the computational box, similarly to the pure advection case. The observed motion confirms that the packet is transported with the velocity imposed by the external velocity field.

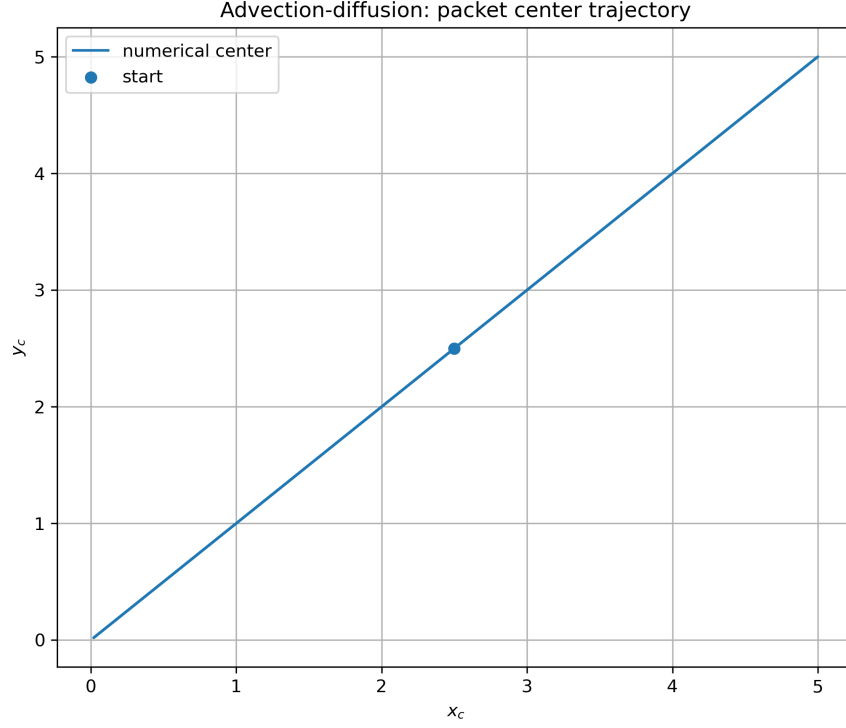


Figure 4: Trajectory of the packet center for advection–diffusion.

The time dependence of the center coordinates is presented in Figure 5. The numerical results follow the theoretical prediction very closely. The apparent discontinuity visible for the  $y$  coordinate is a consequence of the periodic boundary conditions. When the packet reaches the upper boundary of the computational box, it reappears at the lower boundary, producing a jump in the coordinate representation while the physical motion remains continuous.

Therefore, the numerical results indicate that the velocity of the packet center is equal to the velocity field vector  $(1, 1)$ .

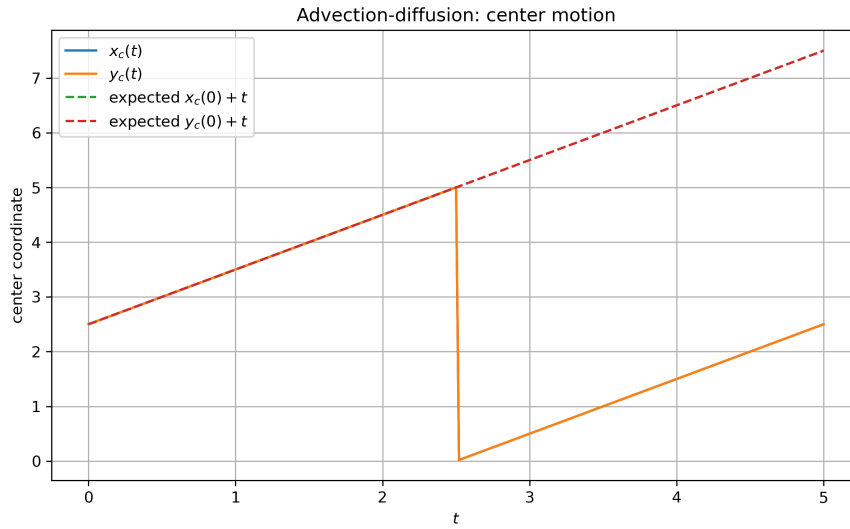


Figure 5: Coordinates of the packet center as functions of time for the advection–diffusion case.

### 3 Conclusions

In this project, the two-dimensional advection–diffusion equation with periodic boundary conditions was solved using the finite element method and the Crank–Nicolson time integration scheme.

For pure advection, the Gaussian packet was transported across the computational domain with a velocity consistent with the imposed velocity field. The packet returned to its initial position due to the periodic boundary conditions, while its shape remained nearly unchanged, showing only a small numerical deformation.

For pure diffusion, the solution became progressively more uniform with time. The maximum value of the distribution decreased, while the minimum value increased, demonstrating the smoothing effect of diffusion.

For the combined advection–diffusion case, the packet was simultaneously transported and broadened. The motion of the packet center followed the theoretical trajectory corresponding to the velocity field  $(1, 1)$ , confirming that the propagation velocity of the packet is equal to the prescribed advection velocity.

Overall, the numerical results reproduced the expected physical behavior of advection, diffusion, and their combined action, demonstrating the correctness of the FEM implementation.